Product Proposal & Executive PRD

Executive product framing, vision, users, scope and differentiation

Document Control

Field	Value
Product	QuickFlow
Document	Product Proposal & Executive PRD
Version	1.0
Status	Proposed / Planning
Audience	Manager, Product/Engineering Review
Source Basis	Supplied project PDF + collected QuickFlow project notes

1. Executive Summary

QuickFlow is an intelligent order fulfillment and scheduling platform for e-commerce operations. Its purpose is to make operational decisions around an order rather than merely record and forward it.

For every order, QuickFlow evaluates inventory across warehouses, selects a suitable fulfillment location, safely reserves inventory under concurrent requests, decomposes the order into fulfillment tasks, schedules those tasks, assigns available workers, manages deadlines and failures, and exposes the resulting decisions through customer and operations interfaces.

The central design decision is to make the **C++ engine the decision-making core**, while React provides the experience layer, Node.js/Express provides the API and integration layer, MongoDB stores persistent state, and WebSocket/Socket.IO provides real-time status propagation.

2. Product Vision

Vision: Build an operations platform that can answer: *“Given current orders, inventory, warehouses, workers and deadlines, what is the most effective way to fulfill these orders?”*

Principle	What it means in QuickFlow
Intelligent	Makes warehouse, priority and scheduling decisions using operational inputs.
Concurrent	Processes independent fulfillment work in parallel while protecting shared state.
Explainable	Shows why a warehouse, priority or schedule was selected.
Resilient	Retries failed work and moves repeated failures to manual review.

3. Business / Operational Problem

A basic e-commerce workflow becomes difficult when multiple warehouses have different stock, many orders arrive together, worker capacity changes, deadlines approach, or multiple customers attempt to reserve the same limited item. QuickFlow is intended to manage these interactions as one coordinated fulfillment problem.

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4. Product Goals

- Reduce inefficient warehouse allocation by considering inventory, workload, processing time, distance and shipping cost.
- Prevent incorrect inventory allocation during concurrent order requests.
- Prioritize work based on order priority and time-to-deadline.
- Increase throughput through parallel task execution and a reusable thread pool.
- Provide operational transparency through dashboards, events and decision explanations.
- Allow managers to evaluate future workload configurations through simulation.

5. Users & Product Surfaces

User	Primary Surface	Core Need
Customer	Customer E-commerce App	Browse, buy and track fulfillment progress.
Operations Manager	Admin Operations Console	Monitor orders, warehouses, workers, queues and system health.
Warehouse Manager	Operations Views	Understand assigned work, worker capacity and inventory state.
Admin	Full Console	Control, analyze and simulate the overall operation.

6. Product Experience Split

Customer experience

Premium, simple e-commerce experience. Customers see order confirmation, fulfillment progress, estimated completion and meaningful activity updates — not internal implementation details such as mutexes, queues or worker IDs.

Operations experience

Dense, technical operations console. Managers see queues, warehouse utilization, workers, decisions, events, analytics and simulation controls.

7. Core Value Proposition

QuickFlow converts fulfillment from a static transaction pipeline into an explainable decision system. It does not only answer “what happened?”; it is designed to answer “why was this decision made?” and “what happens if operating conditions change?”

8. Product Scope

The initial scope includes customer ordering and tracking, multi-warehouse inventory, intelligent allocation, task generation, scheduling strategies, worker management, concurrency controls, failure recovery, order state management, operations dashboards, analytics, event logging and simulation.

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9. High-Level Product Flow

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Customer → React → Node/Express → C++ Engine → Inventory/Warehouse/Scheduler → Workers →
Fulfillment → Shipping → Customer
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The order lifecycle is coordinated through one shared backend state. The customer application and admin console therefore represent two views of the same fulfillment system rather than two disconnected applications.

10. Why This Product Is Different

Conventional View	QuickFlow View
Order is created and forwarded.	Order is evaluated and operationally planned.
Warehouse is selected manually or statically.	Warehouse is scored against fulfillment factors.
Tasks run in a fixed sequence.	Tasks are generated, prioritized and dependency-aware.
Concurrency is an implementation detail.	Concurrency is a first-class correctness requirement.
Dashboard shows counts.	Dashboard explains decisions and operational state.
System handles today's workload.	Simulation evaluates future workload configurations.

11. Signature Feature — Simulation

The simulation engine allows an operations manager to ask questions such as: *“What happens if tomorrow we receive 50,000 orders?”* The manager can vary expected orders, workers, warehouses and scheduling strategy, then compare estimated completion time, late orders and utilization.

Illustrative scenario from the project notes:

Configuration	Estimated Completion	Late Orders
50,000 orders / 30 workers	8.2 hours	1,240
50,000 orders / 40 workers	6.1 hours	380

These figures are **illustrative simulation examples** from the project concept, not fixed production targets. The implemented simulator should calculate results from its actual workload model.

12. MVP Definition

The MVP should prove the complete loop with one reliable path: order creation → inventory check → warehouse selection → safe reservation → task generation → scheduling → worker execution → fulfillment status. Advanced scheduling, real-time updates and simulation can then be layered on top.